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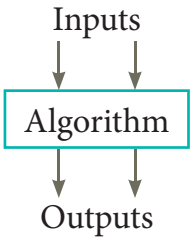

Figure 6.1: Input-output relation

An algorithm is specified by the properties of the given input and the relation between the input and the desired output. In simple words, specification of an algorithm is the desired input-output relation.

The inputs and outputs are passed between an algorithm and the user through variables. The values of the variables when the algorithm starts is known as the initial state, and the values of the variables when the algorithm finishes is known as the final state.

Let P be the required property of the inputs and Q the property of the desired outputs. Then the algorithm S is specified as

1. algorithm_name (inputs)

2. -- inputs : P

3. -- outputs: Q

This specification means that if the algorithm starts with inputs satisfying P, then it will finish with the outputs satisfying Q.

A double dash -- indicates that the rest of the line is a comment. Comments are statements which are used to annotate a program for the human readers and not executed by the computer. Comments at crucial points of flow are useful, and even necessary, to understand the algorithm. In our algorithmic notation, we use double dashes (—) to start a comment line. (In C++, a double slash // indicates that the rest of the line is a comment).

Example 6.6. Write the specification of an algorithm to compute the quotient and remainder after dividing an integer A by another integer B. For example,

divide (22, 5) = 4, 2

divide (15, 3) = 5 , 0

Let A and B be the input variables. We will store the quotient in a variable q and the remainder in a variable r. So q and r are the output variables.

What are the properties of the inputs A and B?

1. A should be an integer. Remainder is meaningful only for integer division, and

2. B should not be 0, since division by 0 is not allowed.

We will specify the properties of the inputs as

— inputs: A is an integer and B ≠ 0

What is the desired relation between the inputs A and B, and the outputs q and r?

1. The two outputs q (quotient) and r (remainder) should satisfy the property

   A = q X B + r, and

2. The remainder r should be less than the divisor B,